

A simple C*-algebra with a unique irreducible representation class

MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint

theorem

Establishes the properties that make the fixed algebra T a counterexample to Naimark's problem.

Statement

Let A be the completed CAR algebra. Let $\pi : A \rightarrow B(\mathcal{H})$ be the direct sum of one chosen GNS representation from each pure-state GNS equivalence class, and let $(L_i)_{i \in I}$ be the unitaries obtained from the fixed CAR homogeneity and shell construction. Put $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$, and write $\iota(a) = \pi(a)$ for the embedding into T . Let $\rho_0 : T \rightarrow B(\mathcal{H})$ be the inclusion representation. Then T is a nonzero, norm-closed, infinite-dimensional unital C*-algebra, ι is injective and unital, and ρ_0 is faithful and nonzero irreducible. Every nonzero irreducible *-representation $\rho : T \rightarrow B(K)$ on a complex Hilbert space K is unitarily equivalent to ρ_0 , even if $\rho(1) = 1$ is not assumed: there is a unitary $U : \mathcal{H} \rightarrow K$ with

$$U\rho_0(t) = \rho(t)U \quad (t \in T).$$

The algebra T is simple: every norm-closed two-sided ideal is $\{0\}$ or T . Moreover, for no complex Hilbert space K is there an injective *-representation $e : T \rightarrow B(K)$ with $e(T) = \mathcal{K}(K)$.

Assumptions

The automorphisms and shell unitaries are obtained from CAR homogeneity and fixed once. No simplicity or uniqueness-of-representation assumption is supplied to the theorem. The Hilbert space K in the representation and compact-operator assertions is arbitrary; separability and preservation of the identity by the given representation are not assumed.

Conclusion

Thus T has exactly one unitary-equivalence class of nonzero irreducible *-representations, but is not isomorphic to the full algebra of compact operators on any Hilbert space. The nonzero, norm-closed algebra T , its injective unital embedding ι and its faithful irreducible inclusion ρ_0 are the same throughout these assertions.

In the displayed inclusion representation, one also has $T \cap \mathcal{K}(\mathcal{H}) = \{0\}$. This follows from the stated properties: the intersection is a closed two-sided ideal of T ; if it were nonzero, simplicity would give $T \subseteq \mathcal{K}(\mathcal{H})$. Then $I_{\mathcal{H}}$ would be compact, so $\mathcal{H}$, and hence $T \subseteq B(\mathcal{H})$, would be finite dimensional, a contradiction. This is a consequence of the theorem, not an extra field of its Lean statement.

Proof route

Apply the cited theorem for a fixed shell family to the family supplied by CAR homogeneity. The construction and the representation-classification theorem give the embedding and unitary equivalences; their cited consequences give simplicity and the exclusion of an algebra of compact operators.

▼ Proof steps

1. Let o denote the class of the distinguished pure state, let f_n be the decreasing root projections, and write $p_n = f_n - f_{n+1}$ for their shells. At o choose $\alpha_o = \text{id}$ and $w_{o,n} = p_n$. In each other class, CAR homogeneity supplies one automorphism α_i , and exact shell matching supplies all $w_{i,n}$ for that same α_i . These choices are made once for the entire construction.
2. The shell construction supplies an injective unital embedding ι and an irreducible inclusion ρ_0 . Since ι embeds the infinite-dimensional algebra A , the algebra T is infinite dimensional. Its norm closure is part of its construction, and its nonzeroness follows from injectivity of ι .
3. The representation-classification theorem for this fixed family gives unitary equivalence with ρ_0 for every unital nonzero irreducible representation. The cited unitality lemma gives $\rho(1) = I_K$ for every nonzero irreducible $*$ -representation ρ of T , so the same conclusion applies without initially assuming that ρ preserves the identity.
4. The cited simplicity theorem applies because ρ_0 is faithful and all nonzero irreducible representations are unitarily equivalent to it. The cited compact-operator exclusion applies because T is unital and infinite dimensional. The Lean theorem collects these proved properties for the same algebra.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint`
- The abbreviation for the full structural assertion —
`MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleEndpoint`
- The ten properties in the exact statement — `MathlibAnnex.CStarAlgebra.CAR.ShellFamilyEndpoint`
- The structural theorem for a fixed shell family —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint`
- The once-chosen automorphisms and shell partial isometries —
`MathlibAnnex.CStarAlgebra.CAR.representativeShellDataOfHomogeneity`
- Unitality forced by nonzero irreducibility —
`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalRepresentation.map_one_eq_one_of_isIrreducible`
- Simplicity of the constructed algebra —
`MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget`
- Infinite dimensionality of the constructed algebra —
`MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional_shellFamilyTarget`

Supporting route explanation

Lean source signature (exact)

```
theorem atomicCounterexampleEndpoint : AtomicCounterexampleEndpoint.{v}
```

In the source	Mathematical meaning
theorem atomicCounterexampleEndpoint	This theorem has no shell-family or simplicity premise. It proves the stated properties for the fixed homogeneity construction.
AtomicCounterexampleEndpoint.{v}	The result abbreviation is ShellFamilyEndpoint.{v} homogeneityShellFamily. The two separate complete exact declarations and ten independent field rows below resolve the short result type locally.
homogeneityShellFamily	The once-fixed automorphisms and shell links supplied by proved CAR homogeneity. They determine the same target T , source $\iota : A \rightarrow T$ and inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ across every field, before any comparison Hilbert space K is quantified.

These are two separate exact related declarations: the result abbreviation (AtomicCounterexample.lean, lines 92–93), then its ten-field proposition (AtomicEndpoint.lean, lines 194–229). Neither is a replacement or extension of the theorem signature above. Here the family is the fixed homogeneityShellFamily, so T, ι, ρ_0 have exactly the meaning given in this Card.

```
abbrev AtomicCounterexampleEndpoint : Prop :=
  ShellFamilyEndpoint.{v} homogeneityShellFamily
```

```
structure ShellFamilyEndpoint (family : RepresentativeShellFamily) : Prop where
  nontrivial_target : Nontrivial (ShellFamilyTarget family)
  isClosed_target :
    IsClosed
      (ShellFamilyTarget family : Set
        (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
          MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState))
  source_injective : Function.Injective (shellFamilySourceHom family)
  source_unital : shellFamilySourceHom family 1 = 1
  not_finiteDimensional_target :
    ¬ FiniteDimensional  $\mathbb{C}$  (ShellFamilyTarget family)
  ambient_injective : Function.Injective (shellFamilyInclusion family)
  isIrreducible_ambient :
    Representation.IsIrreducible (shellFamilyInclusion family)
  captures_nonunital :
    ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
      [CompleteSpace K]
      (rho : NonUnitalRepresentation
        (A := ShellFamilyTarget family) (H := K)),
```

```

rho.IsIrreducible →
  ∃ U :
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState ≈i[ $\mathbb{C}$ ] K,
  ∀ (a : ShellFamilyTarget family)
    (x : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState),
    U (shellFamilyInclusion family a x) = rho a (U x)
closedIdeal_dichotomy :
  ∀ I : TwoSidedIdeal (ShellFamilyTarget family),
    IsClosed (I : Set (ShellFamilyTarget family)) → I = ⊥ ∨ I = T
not_compactOperatorModel :
  ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
    [CompleteSpace K]
    (e : ShellFamilyTarget family →na[ $\mathbb{C}$ ] (K → L[ $\mathbb{C}$ ] K)),
    ¬ (Function.Injective e ∧
      (∀ a : ShellFamilyTarget family, IsCompactOperator (e a)) ∧
      ∀ T : K → L[ $\mathbb{C}$ ] K, IsCompactOperator T →
        ∃ a : ShellFamilyTarget family, e a = T)

```

In the source	Mathematical meaning
nontrivial_target	The fixed algebra T is nonzero.
isClosed_target	The set $T \subseteq B(\mathcal{H})$ is closed for the operator norm.
source_injective	The fixed source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, is injective.
source_unital	The same map sends 1_A to 1_T .
not_finiteDimensional_target	T is not finite-dimensional over $\mathbb{C}$.
ambient_injective	The literal inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ is injective.
isIrreducible_ambient	This same ρ_0 is nonzero irreducible: it has no proper nonzero closed reducing subspace.
captures_nonunital	After the family and T have been fixed, for every complete complex Hilbert space K in universe v and every nonzero irreducible possibly nonunital $\rho : T \rightarrow B(K)$, there is a surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ with $U(\rho_0(t)x) = \rho(t)Ux$ for every $t \in T$, $x \in \mathcal{H}$.
closedIdeal_dichotomy	For every two-sided ideal $\mathfrak{a}$ in T , if its underlying set is norm closed, then $\mathfrak{a} = \{0\}$ or $\mathfrak{a} = T$.
not_compactOperatorModel	For every complete complex Hilbert space K and every possibly nonunital star homomorphism $e : T \rightarrow B(K)$, it is impossible that e is injective, all $e(t)$ are compact, and every compact operator on K equals some $e(t)$. The zero space is included.

Further source notes: `AtomicCounterexampleEndpoint` abbreviates `ShellFamilyEndpoint` `homogeneityShellFamily`; the separate abbreviation and the full structure are linked. In those fields, `atomicCounterexampleRepresentation` is ρ_0 , `atomicCounterexampleSourceHom` is ι , and `captures_nonunital` is the unitary-equivalence assertion for an arbitrary nonzero irreducible representation, without an initial unit-preservation assumption. These names occur in the linked structure and proof, not as extra arguments in this short signature.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint`

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